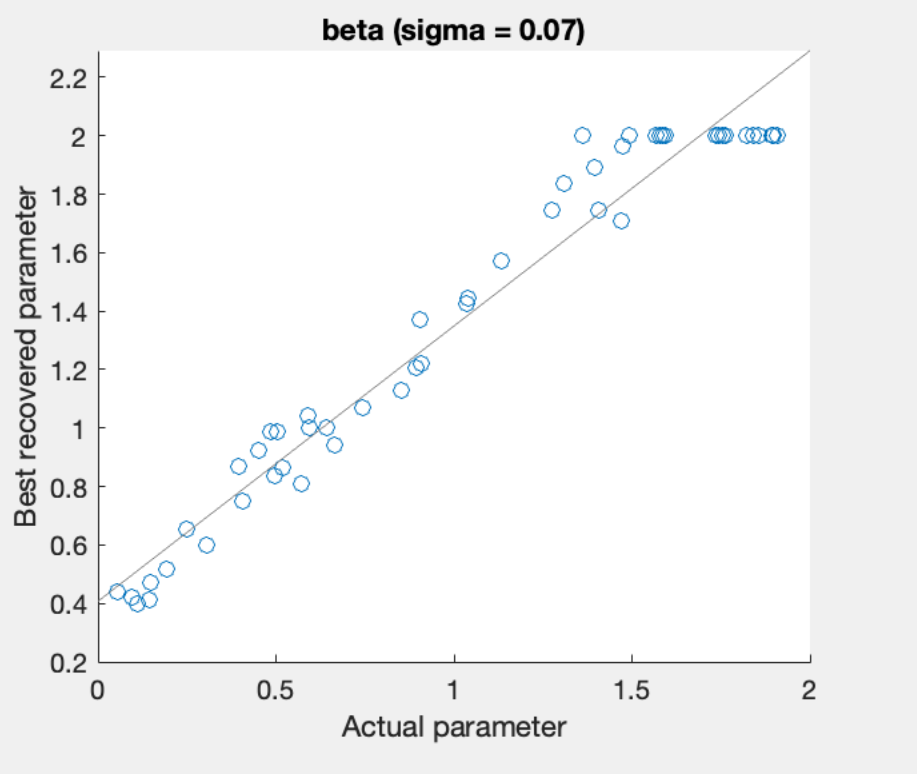
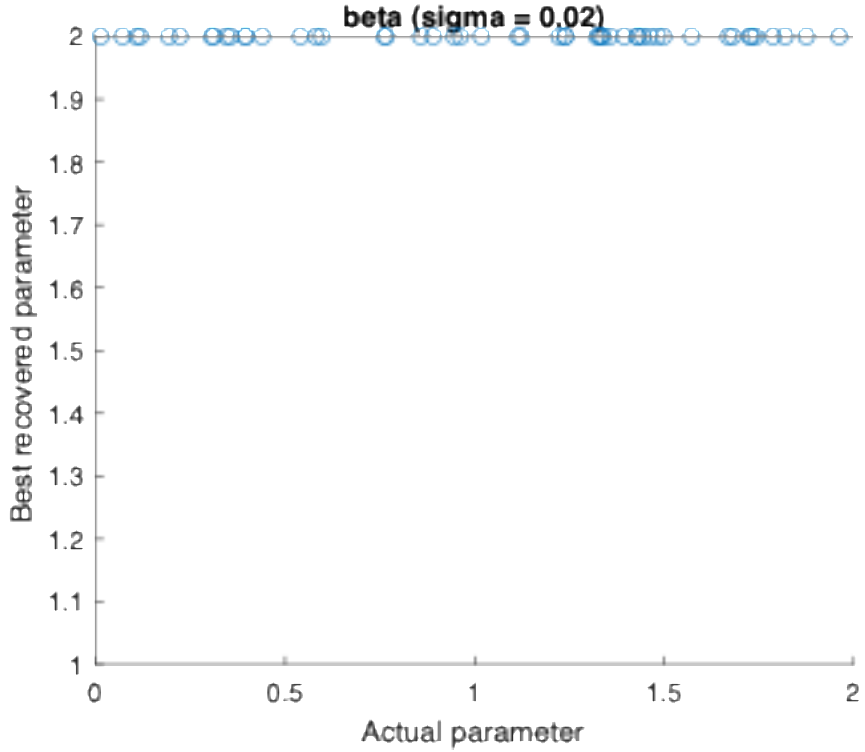
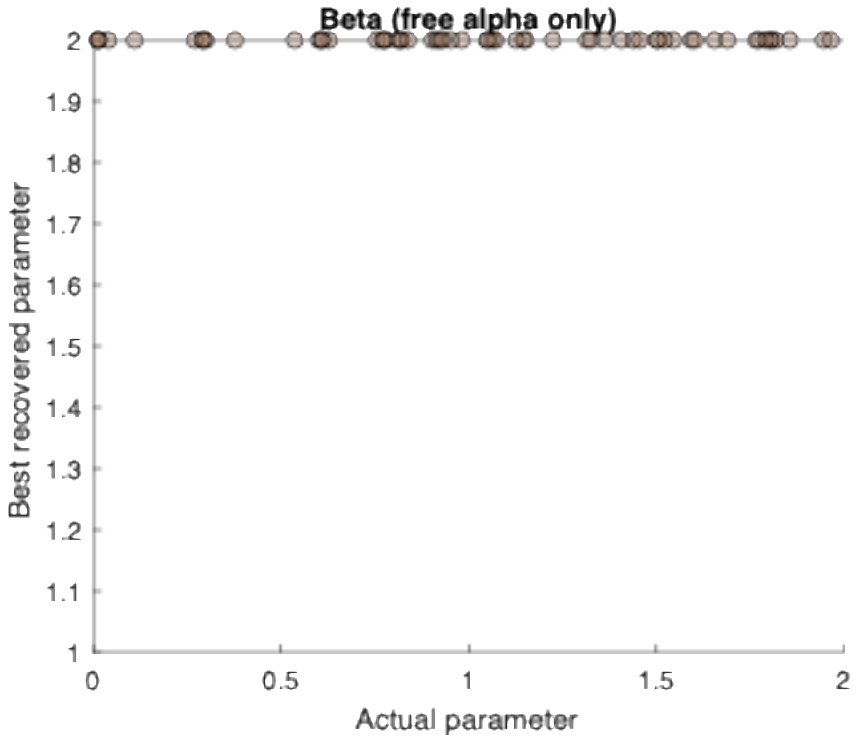
Only beta free and recovered (all fixed but for different levels of sigma)

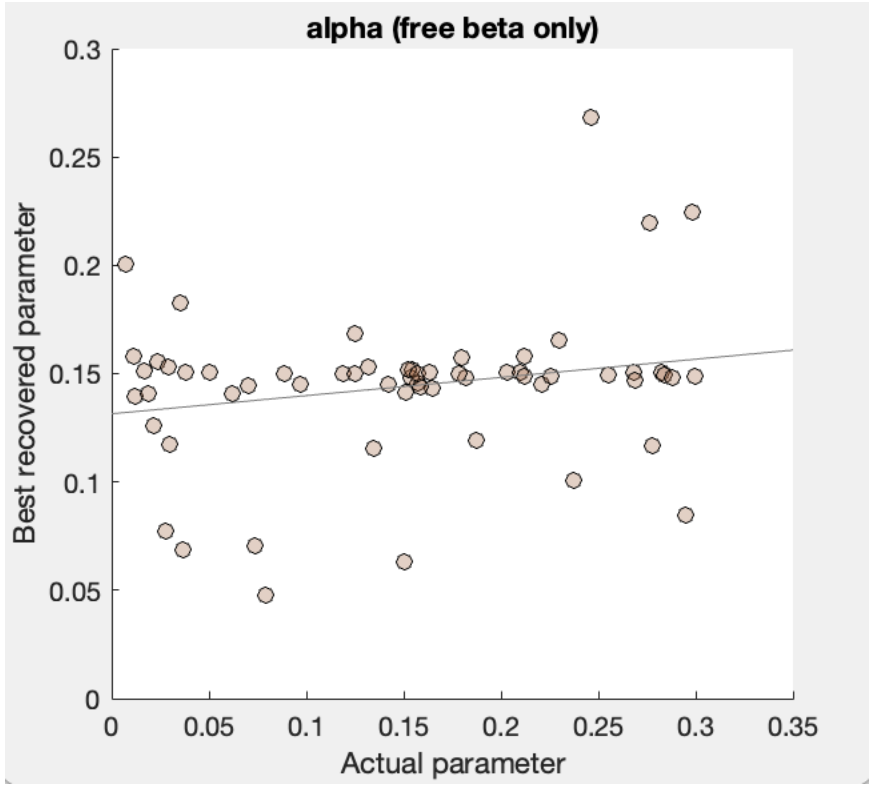
- Recovery better when sigma = 0.07
- Recovery worse when sigma = 0.02



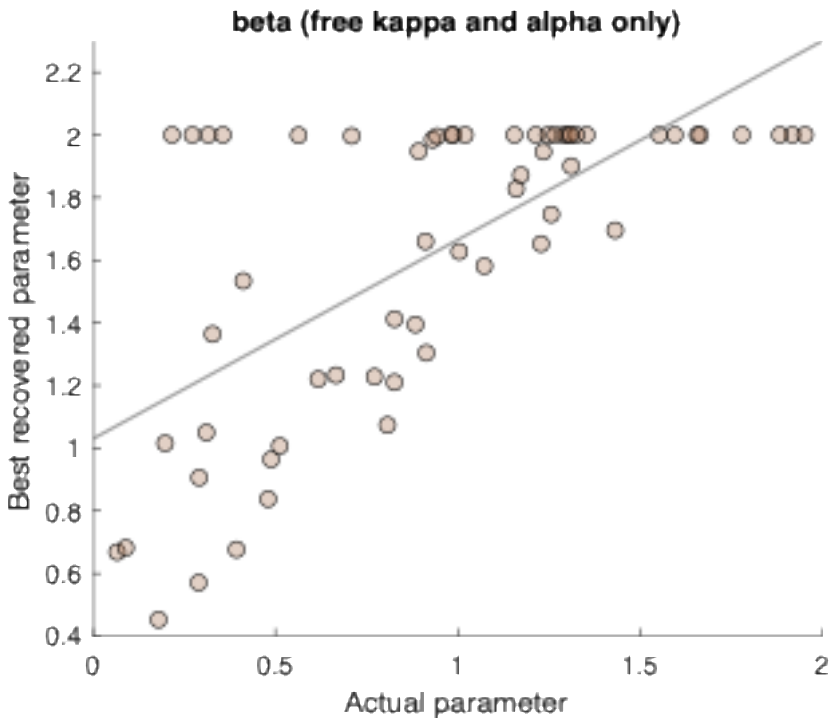


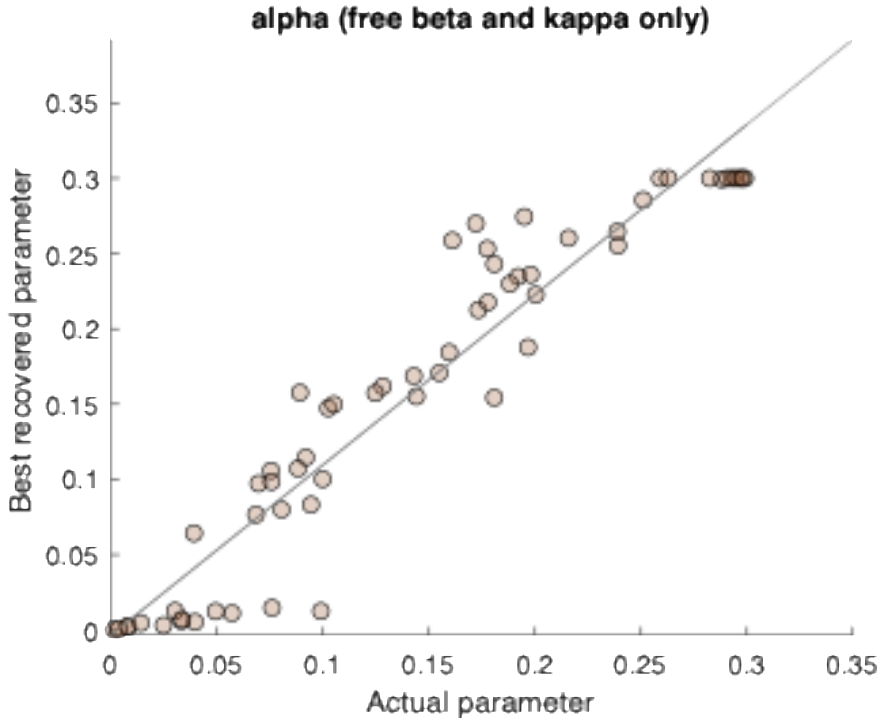
Alpha and beta free and recovered

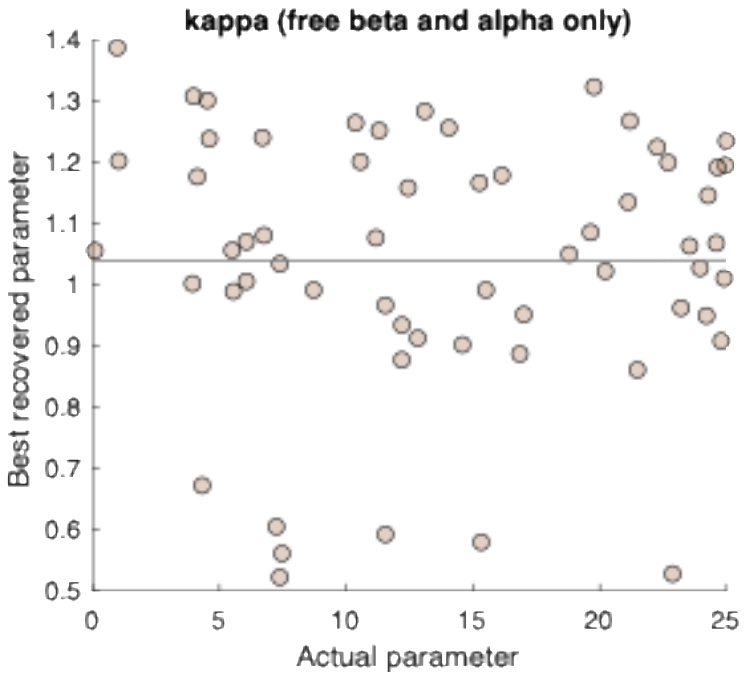




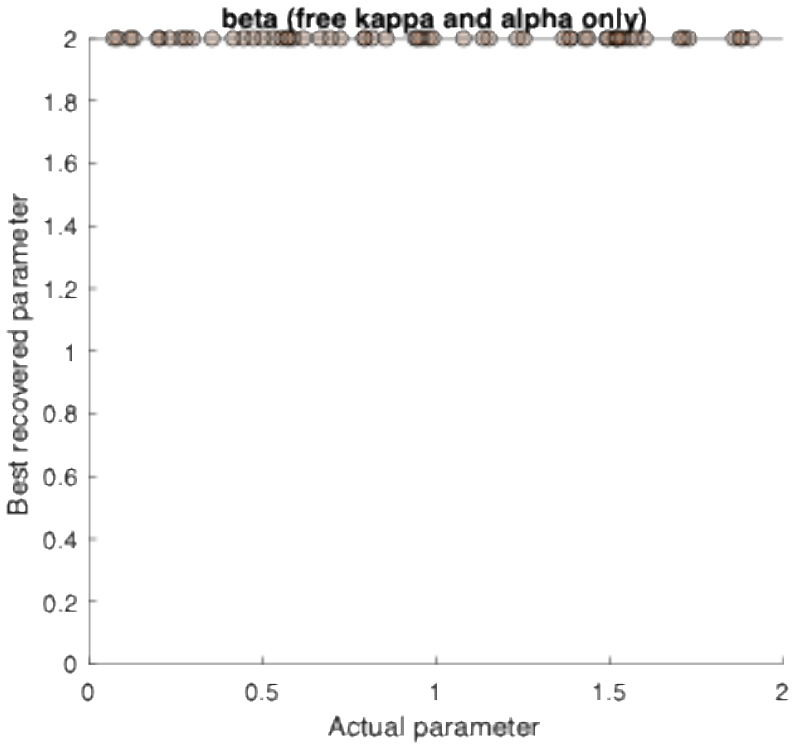
Alpha, beta, kappa free and recovered
(high sigma = 0.07)

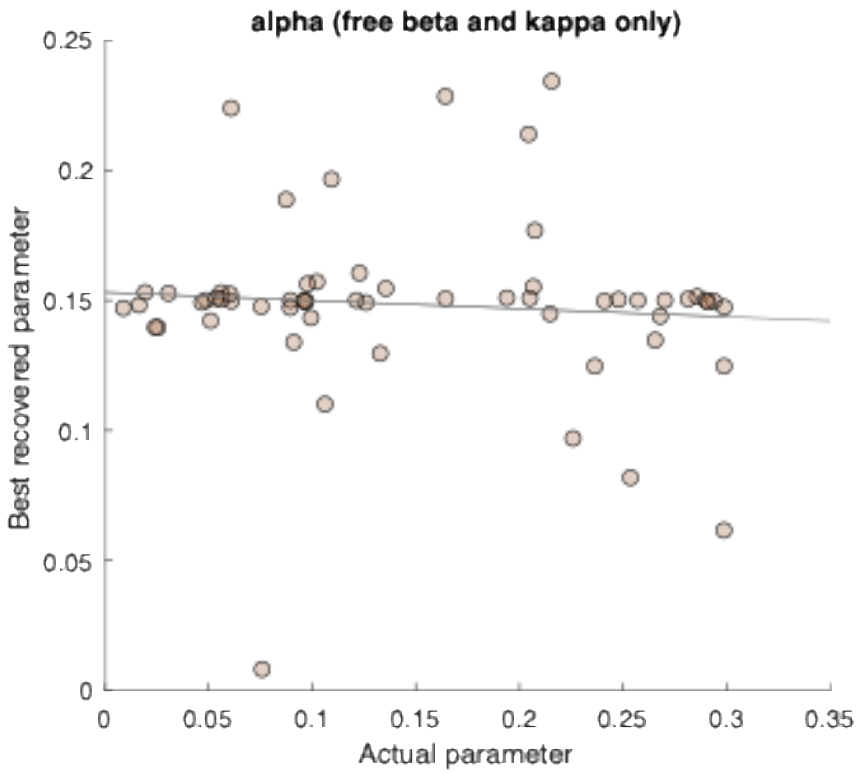


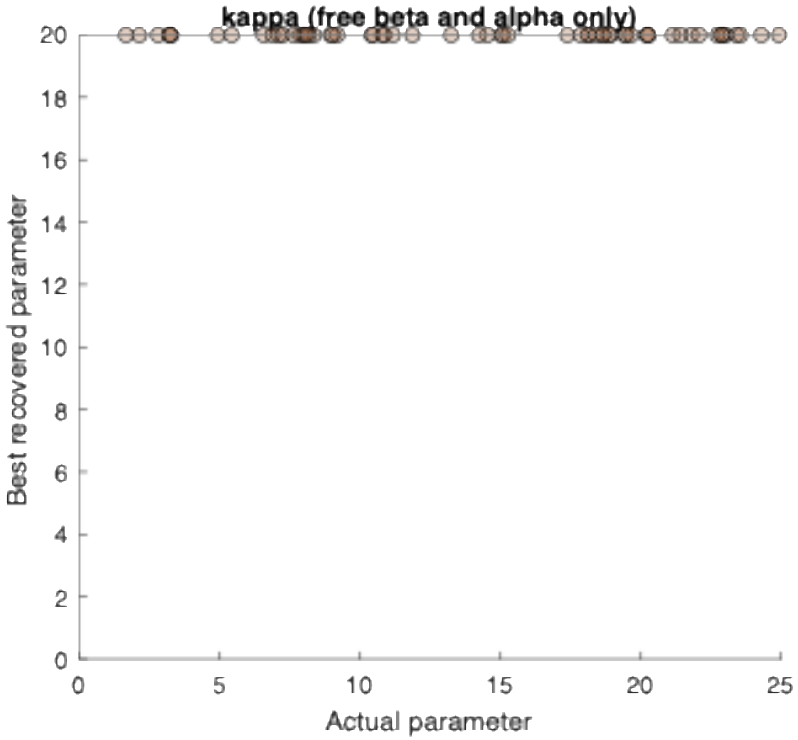




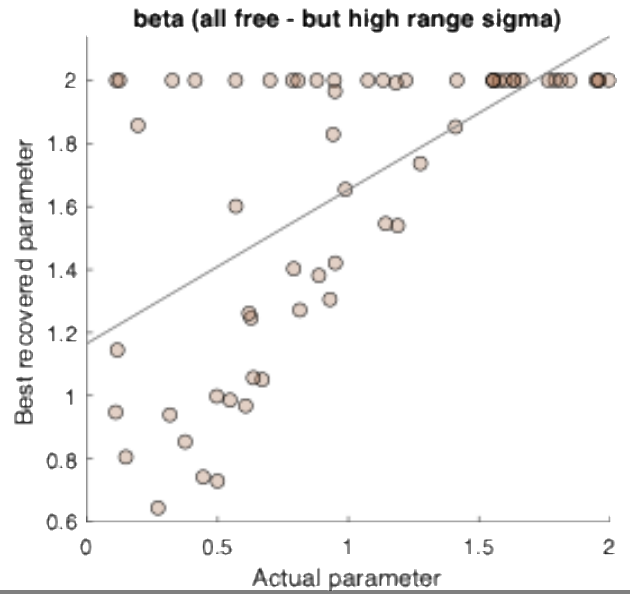
Alpha, beta, kappa free and recovered
(low sigma = 0.02)

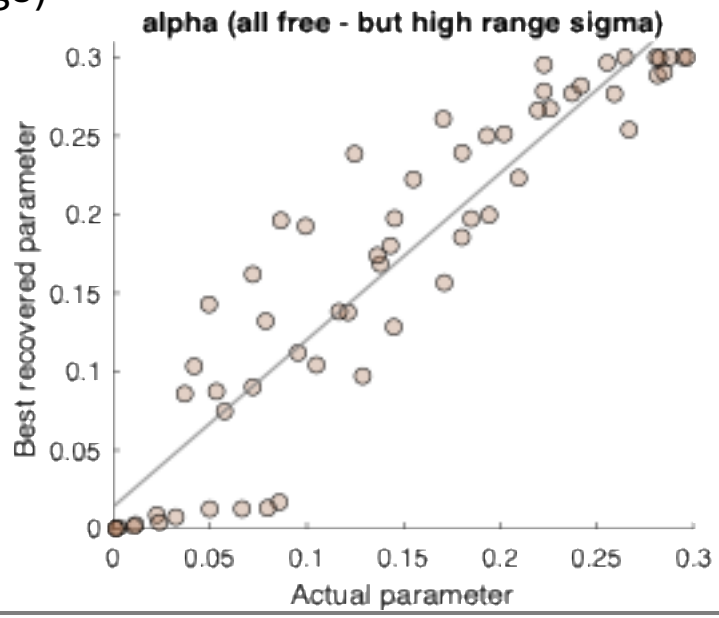


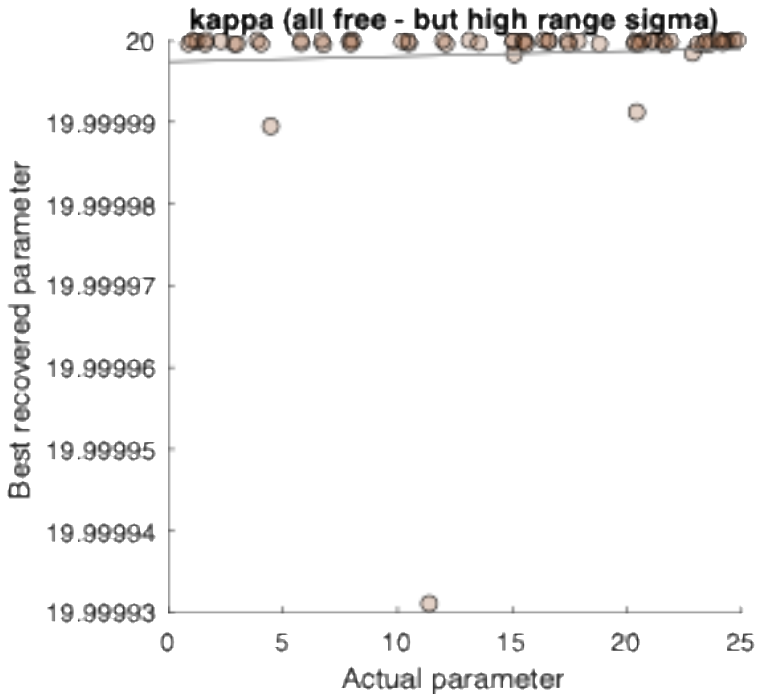


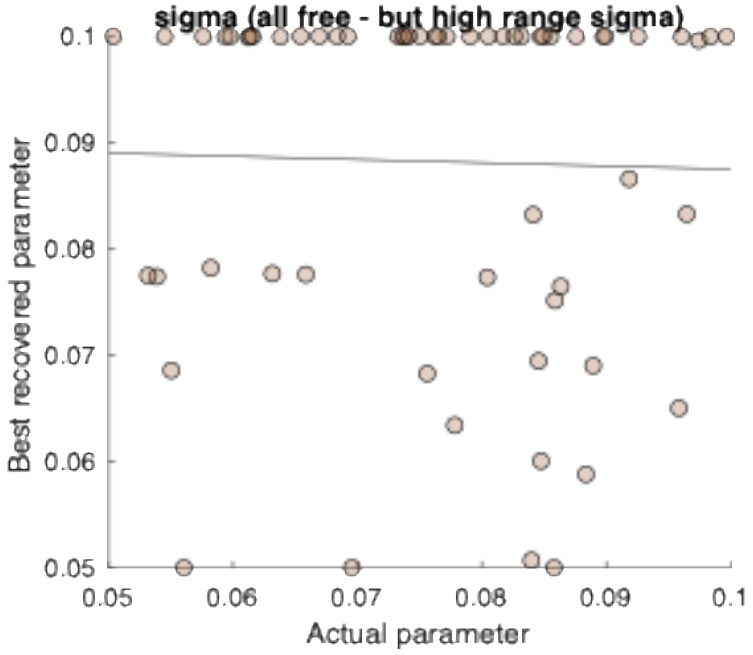


All free and recovered (high sigma range)

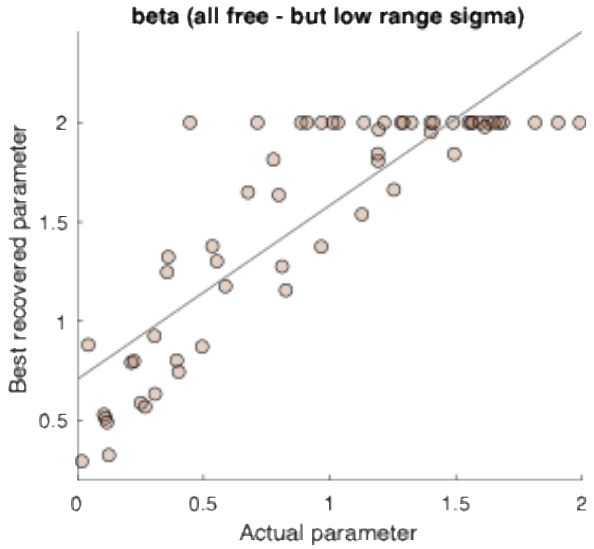


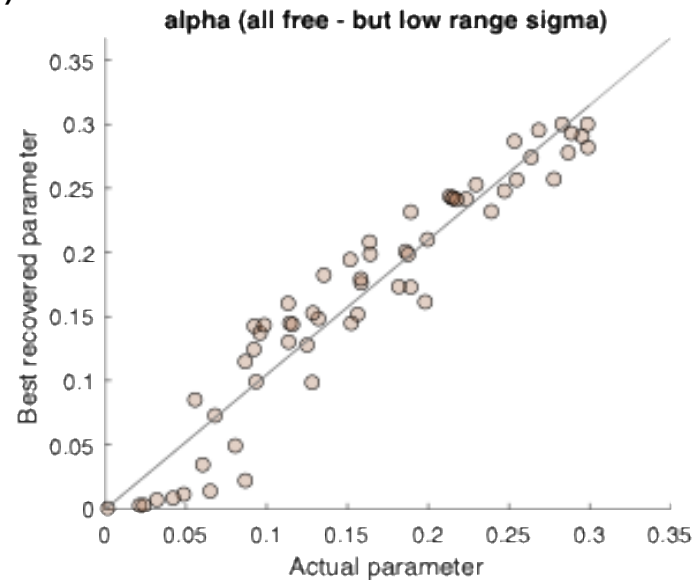


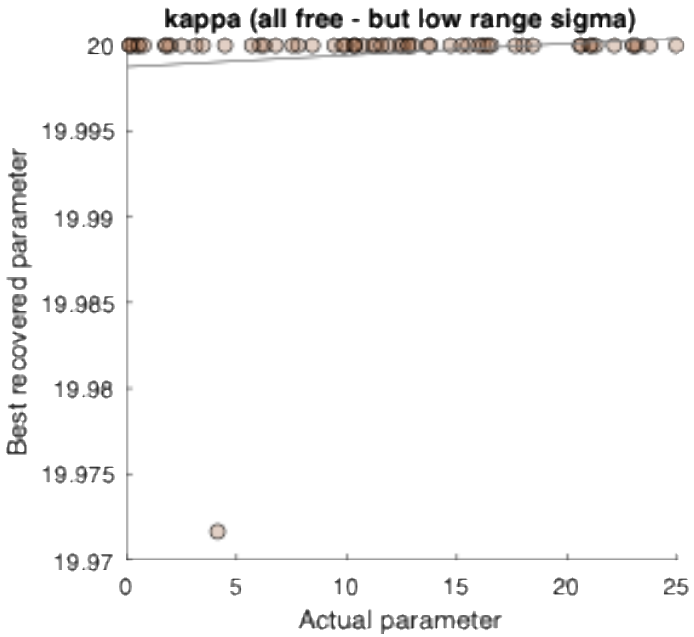


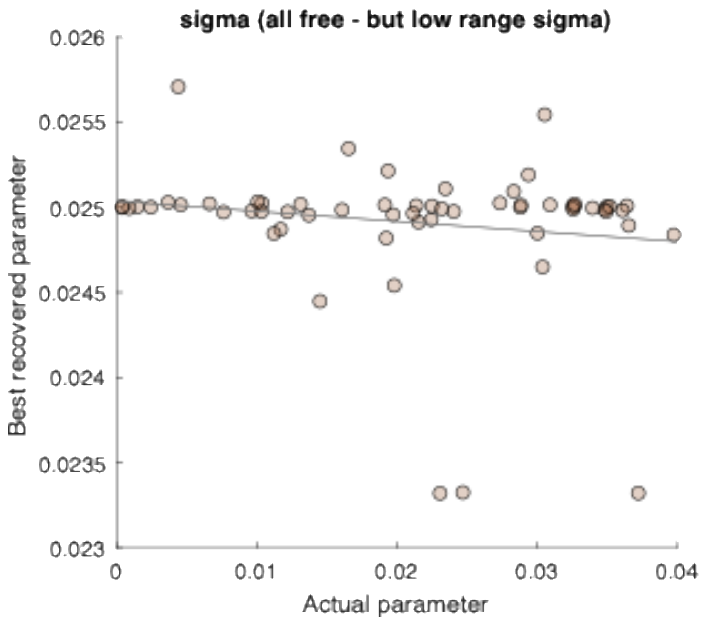


All free and recovered (low sigma range)

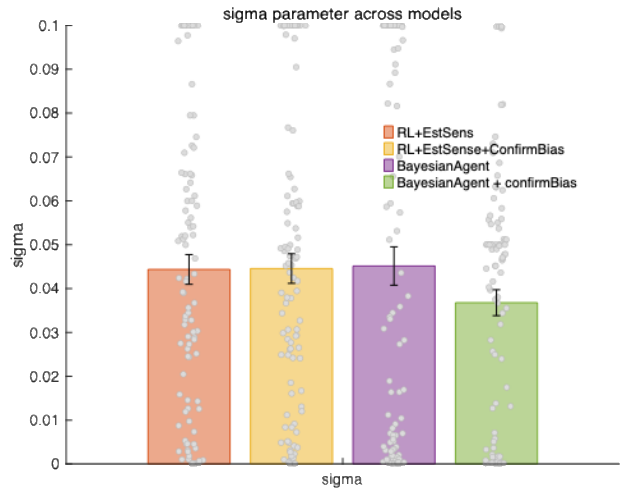


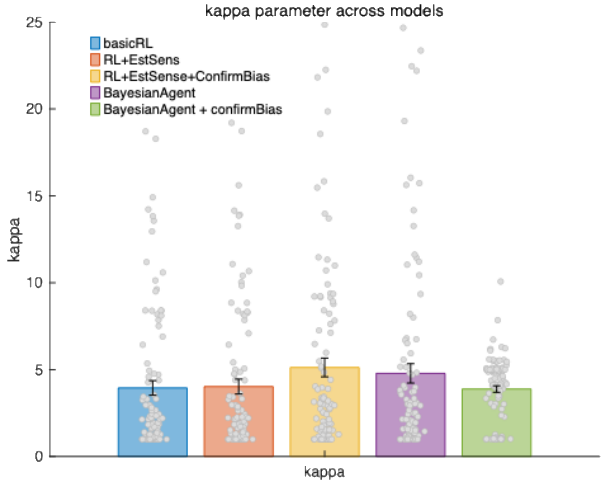


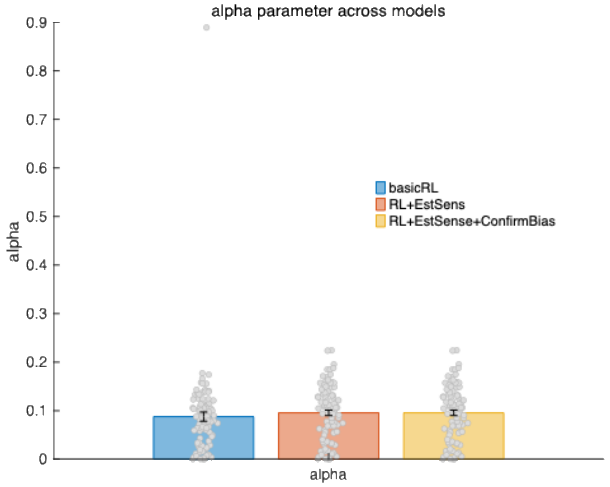


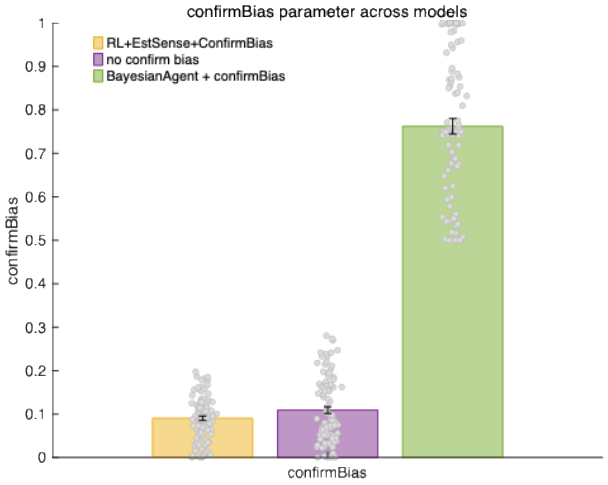


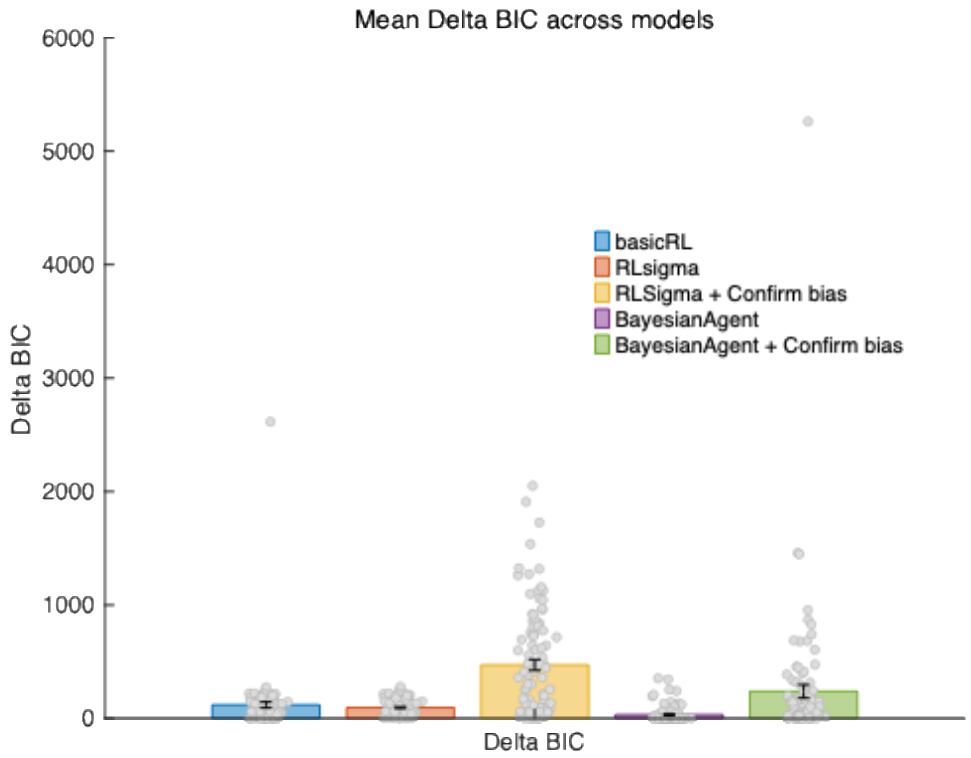
No integration over observations +
No anti-sigmoid transformation

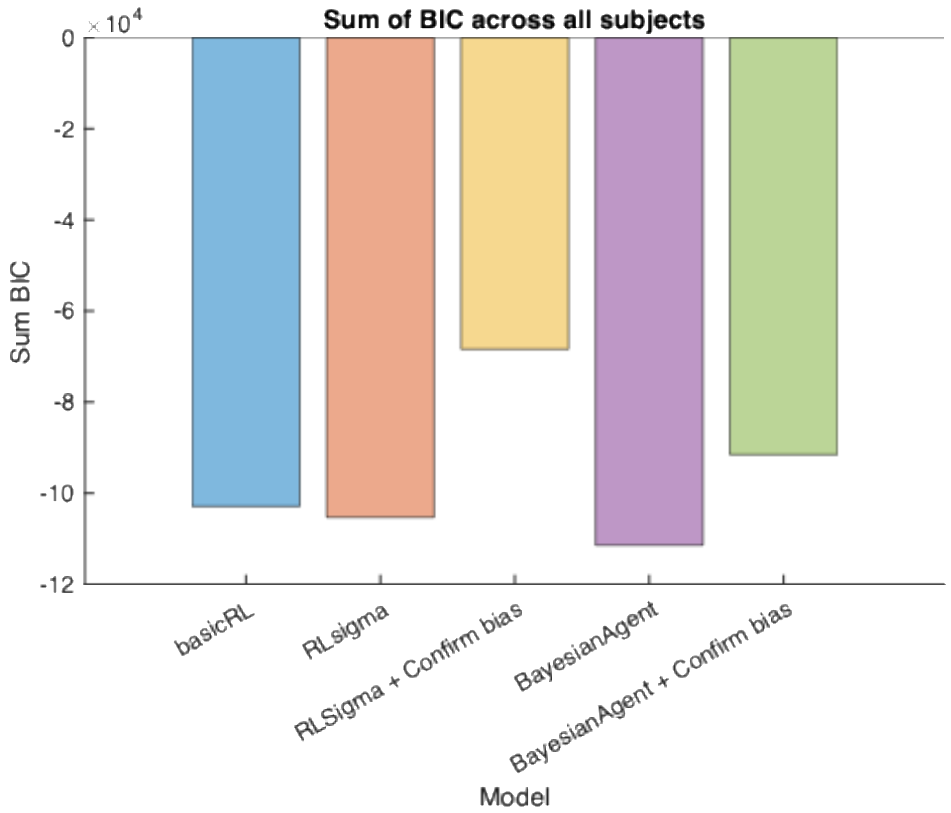

sigma parameter across models
RL+EstSens
RL+EstSense+ConfirmBias
BayesianAgent
BayesianAgent + confirmBias
sigma
sigma
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1

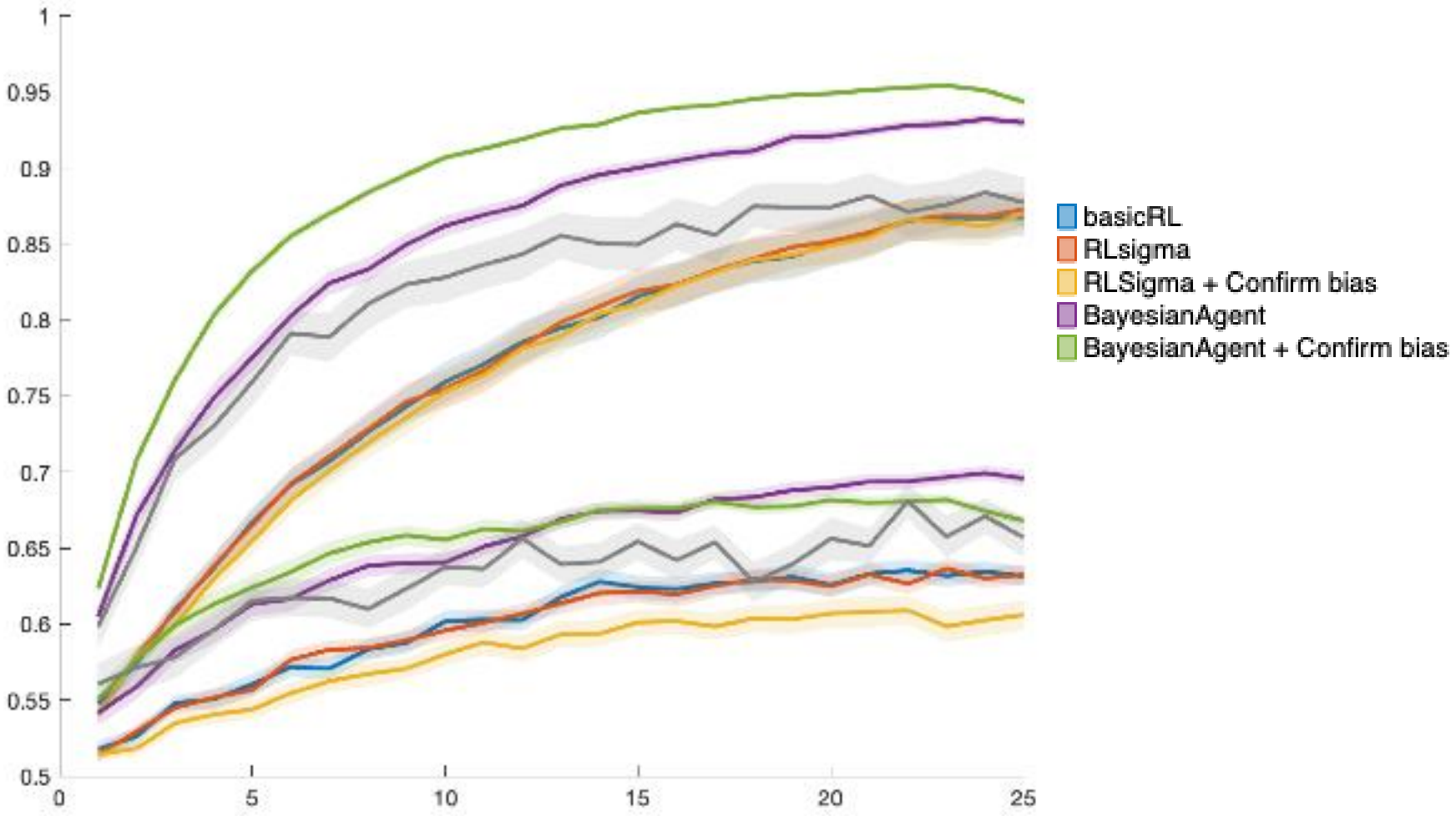

kappa parameter across models
basicRL
RL+EstSens
RL+EstSense+ConfirmBias
BayesianAgent
BayesianAgent + confirmBias
kappa
kappa
0
5
10
15
20
25

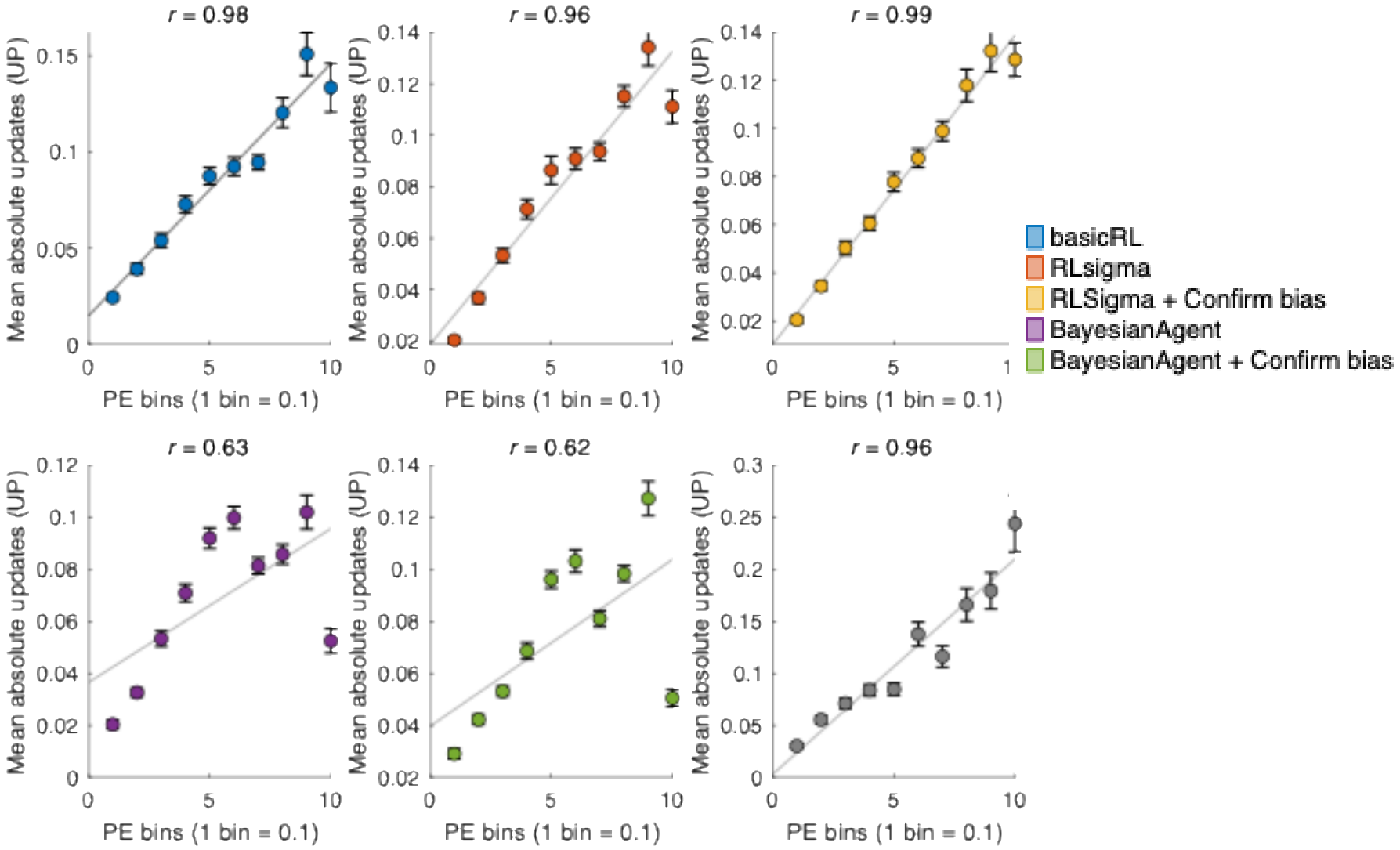

alpha parameter across models
basicRL
RL+EstSens
RL+EstSense+ConfirmBias
alpha
alpha
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9

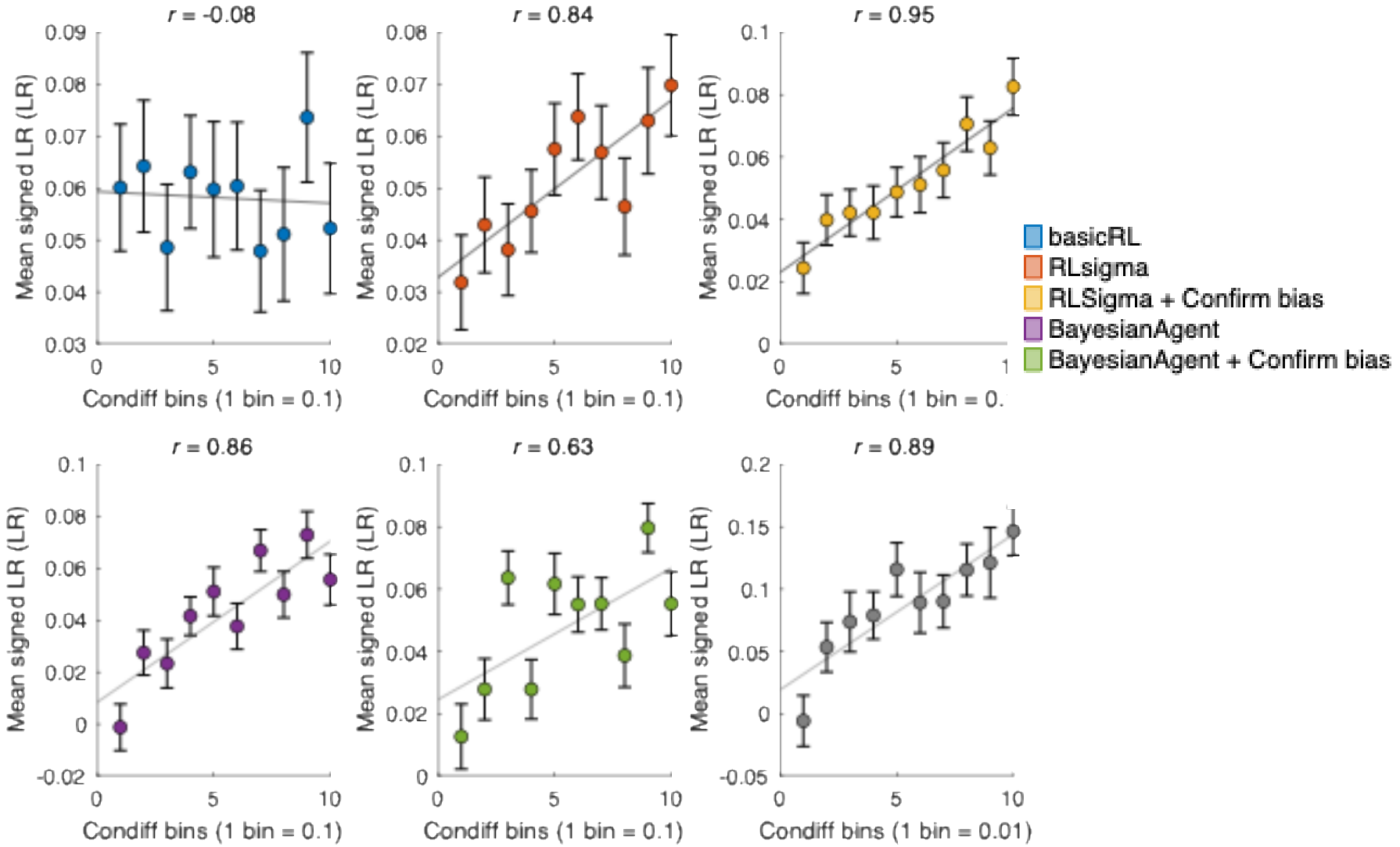

confirmBias parameter across models
RL+EstSense+ConfirmBias
no confirm bias
BayesianAgent + confirmBias
confirmBias
confirmBias
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

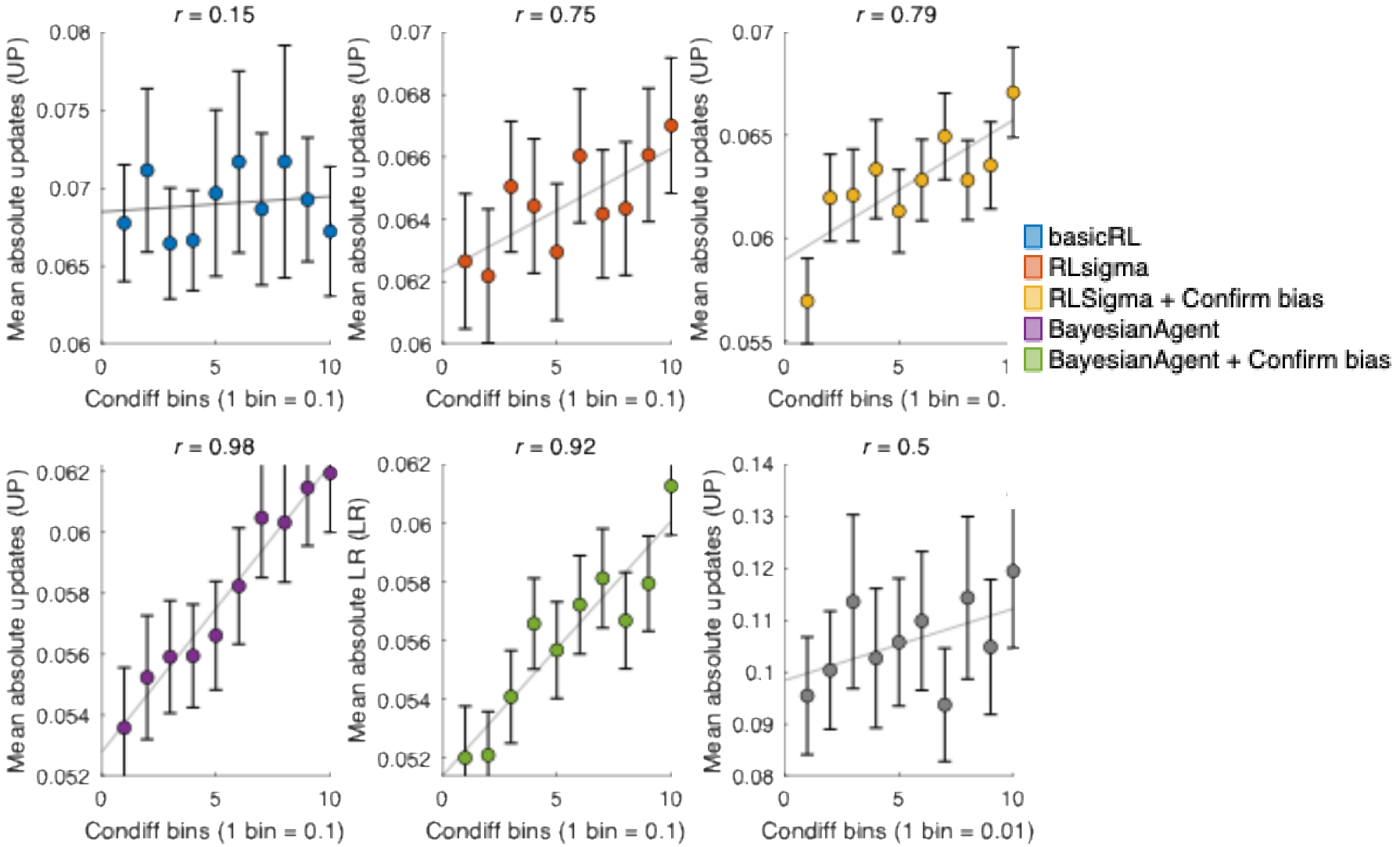

Mean Delta BIC across models
6000
5000
4000
3000
2000
1000
0
Delta BIC
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias
Delta BIC

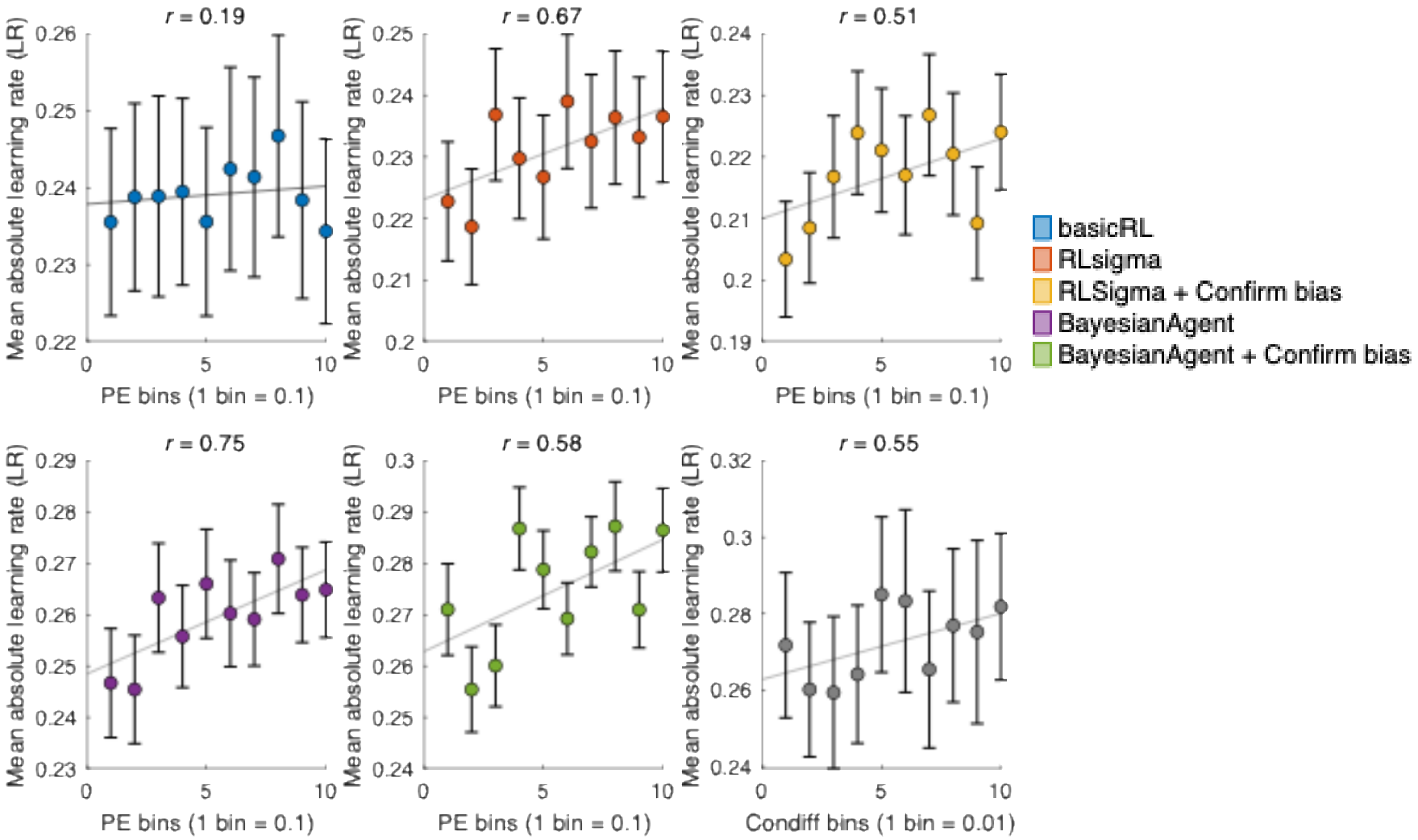

Sum of BIC across all subjects
×10^4
0
-2
-4
-6
-8
-10
-12
Sum BIC
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias
Model


1
0.95
0.9
0.85
0.8
0.75
0.7
0.65
0.6
0.55
0.5
0
5
10
15
20
25
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias

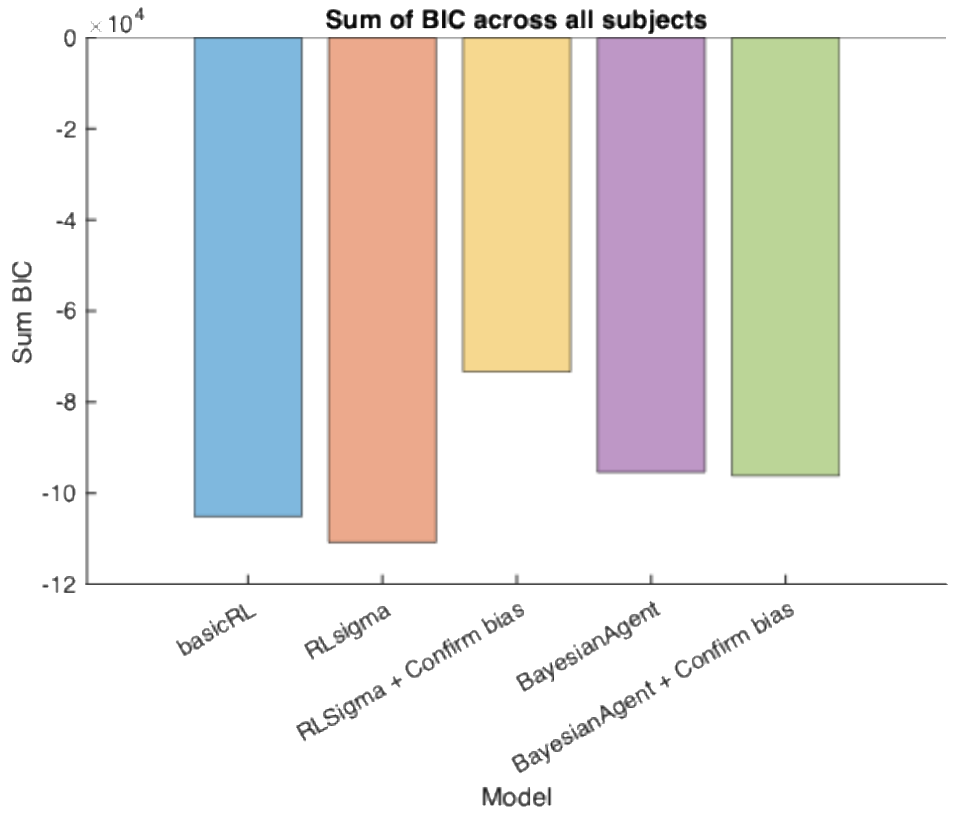

r = 0.98
r = 0.96
r = 0.99
r = 0.63
r = 0.62
r = 0.96
Mean absolute updates (UP)
PE bins (1 bin = 0.1)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias

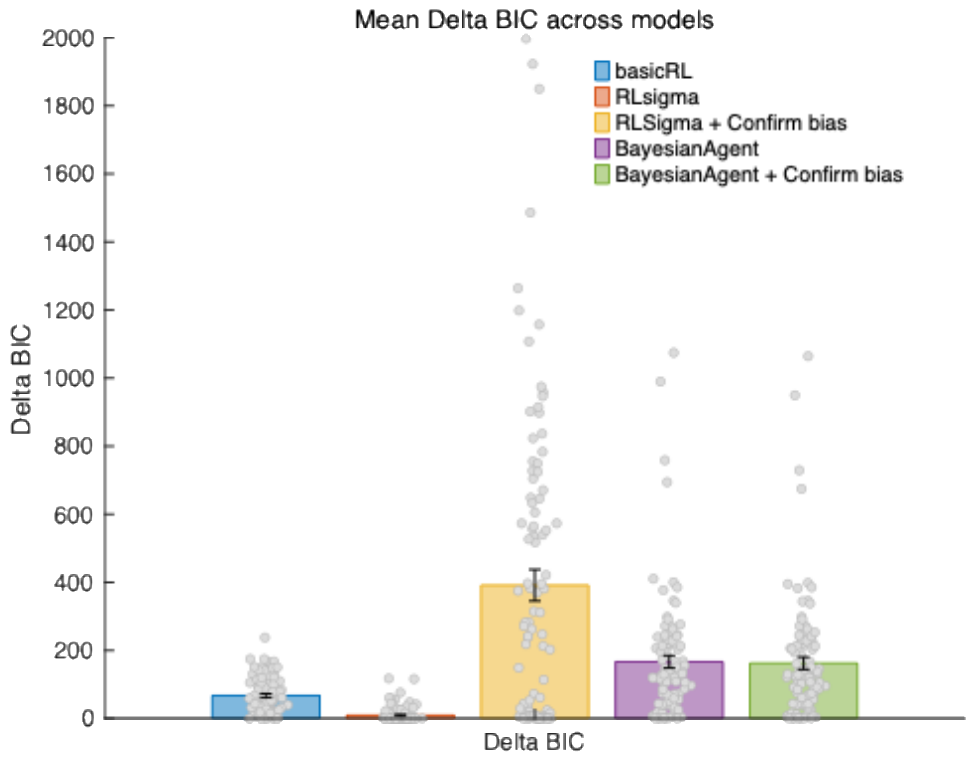

r = -0.08
r = 0.84
r = 0.95
r = 0.86
r = 0.63
r = 0.89
Mean signed LR (LR)
Condiff bins (1 bin = 0.1)
Condiff bins (1 bin = 0.01)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias

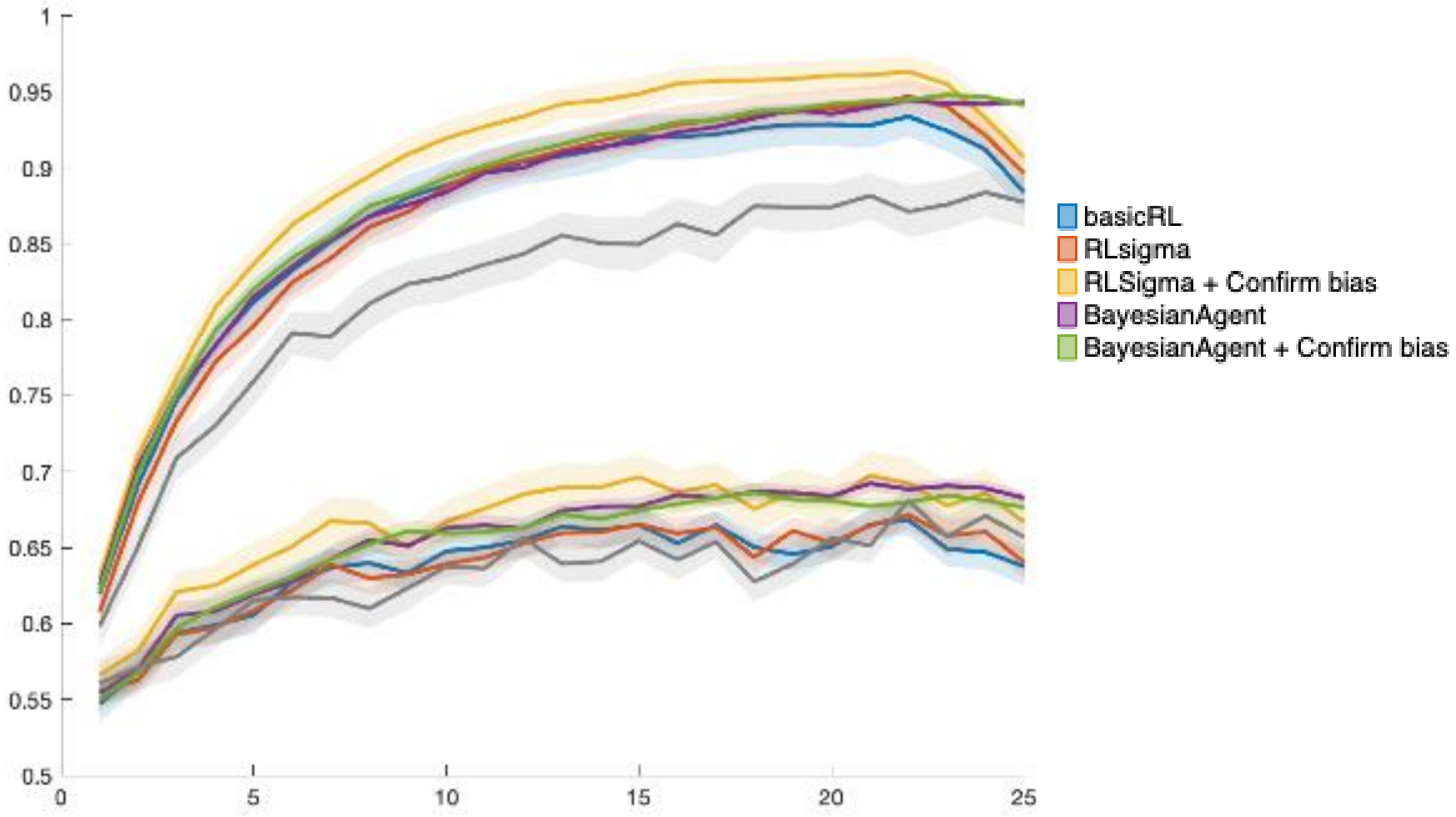

r = 0.15
r = 0.75
r = 0.79
r = 0.98
r = 0.92
r = 0.5
Mean absolute updates (UP)
Mean absolute LR (LR)
Condiff bins (1 bin = 0.1)
Condiff bins (1 bin = 0.01)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias

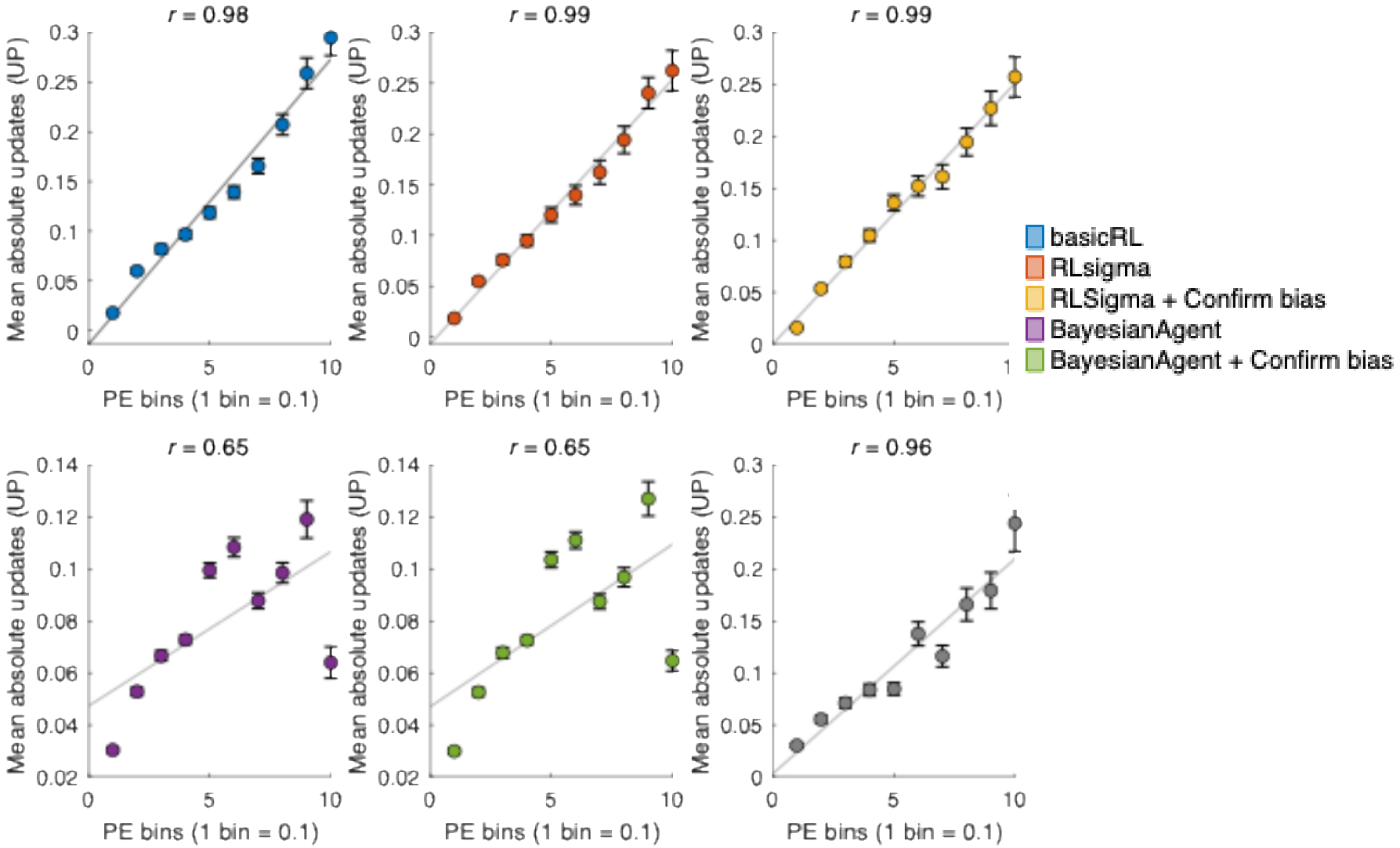

r = 0.19
r = 0.67
r = 0.51
r = 0.75
r = 0.58
r = 0.55
Mean absolute learning rate (LR)
PE bins (1 bin = 0.1)
Condiff bins (1 bin = 0.01)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias

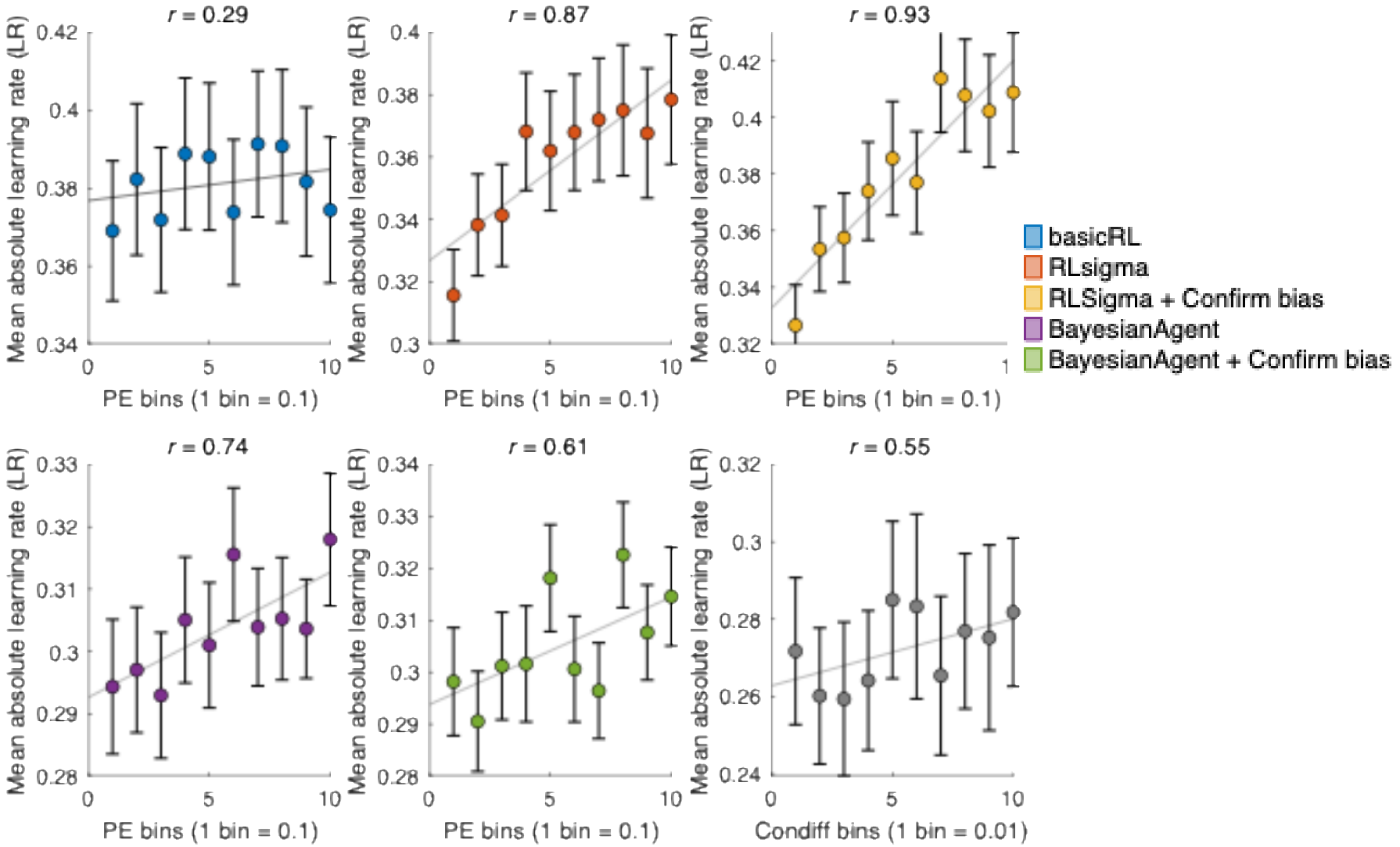
Integration over observations +
No anti-sigmoid transformation

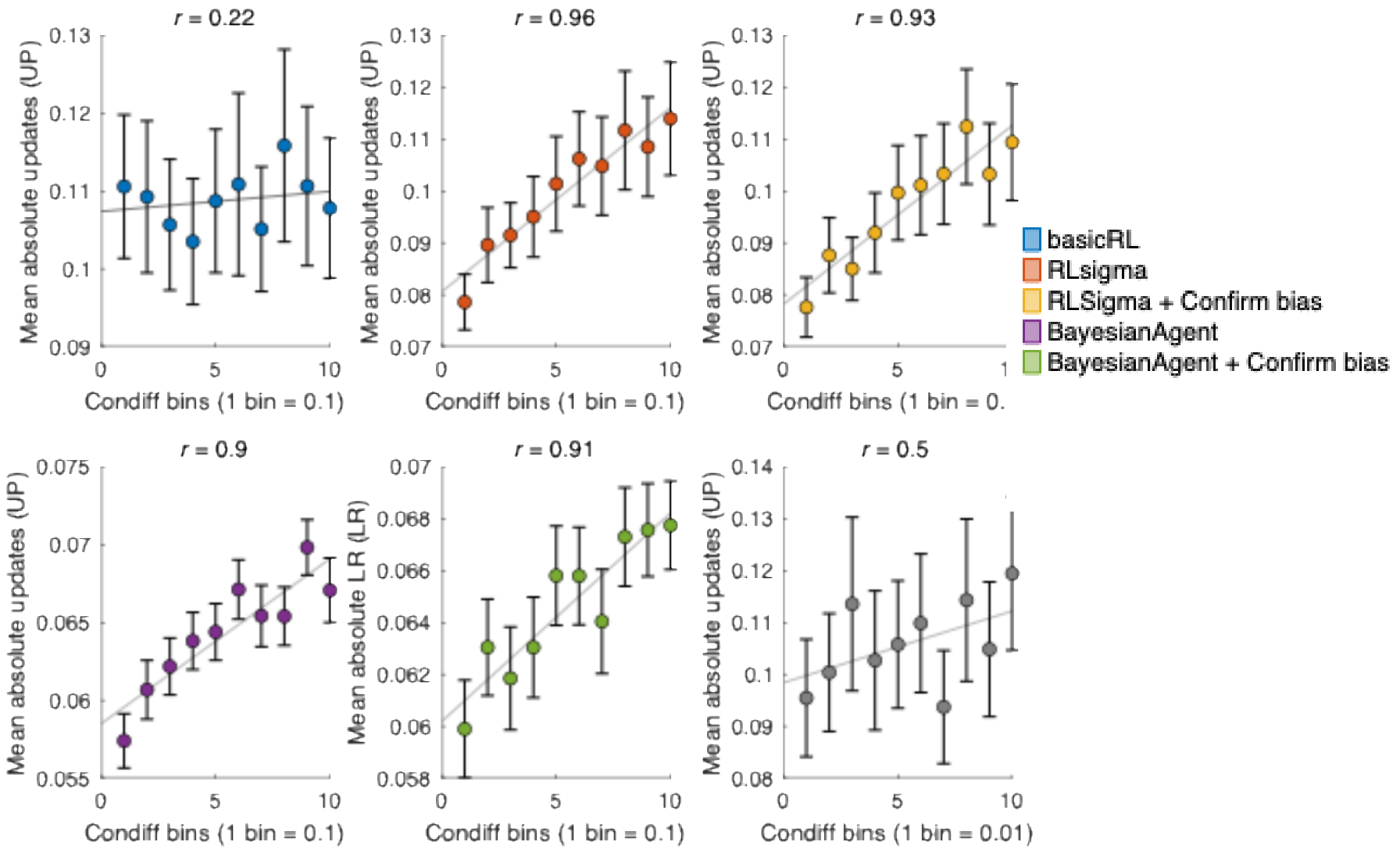

Sum of BIC across all subjects
×10^4
0
-2
-4
-6
-8
-10
-12
Sum BIC
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias
Model

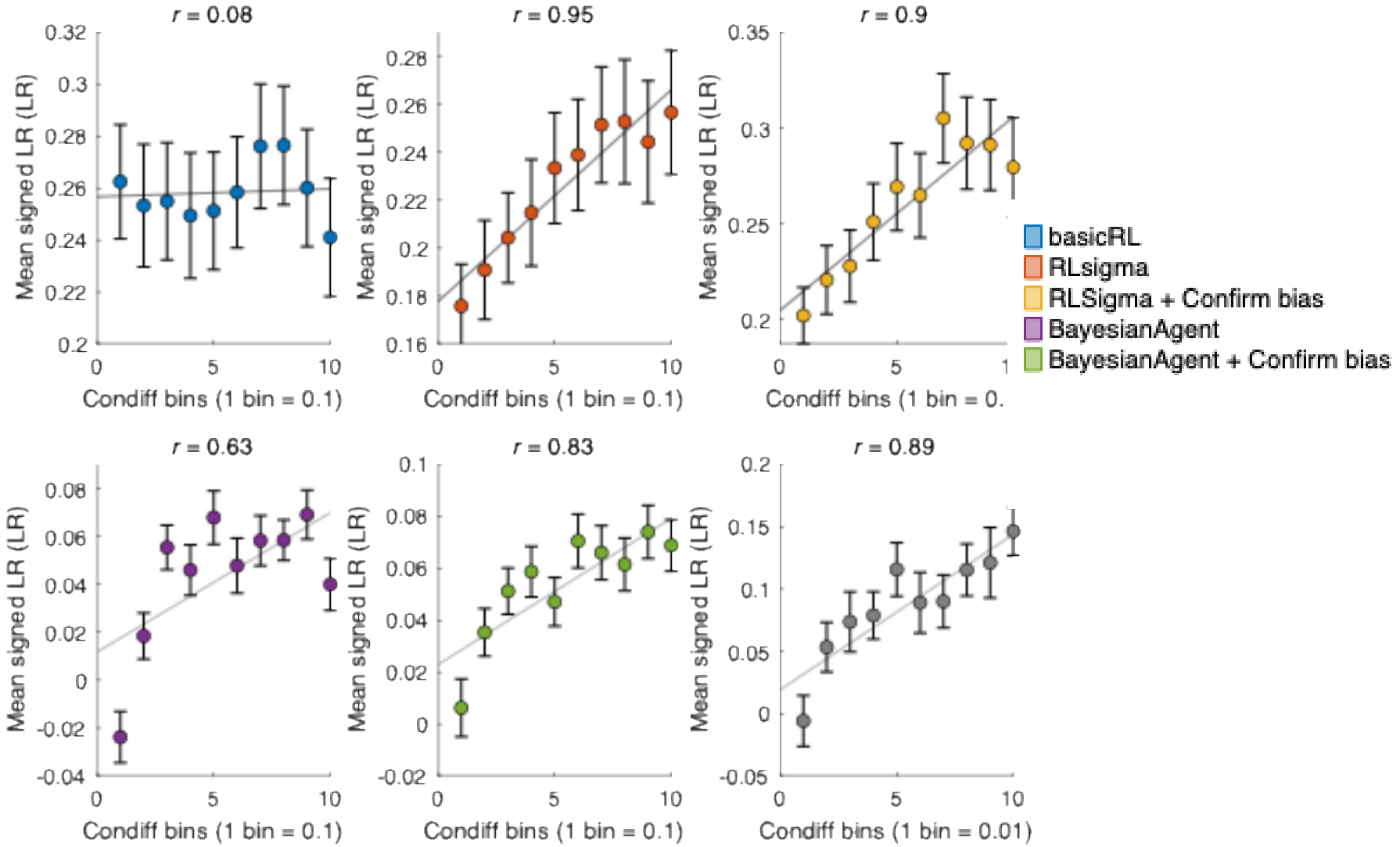

Mean Delta BIC across models
2000
1800
1600
1400
1200
1000
800
600
400
200
0
Delta BIC
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias
Delta BIC


1
0.95
0.9
0.85
0.8
0.75
0.7
0.65
0.6
0.55
0.5
0
5
10
15
20
25
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias


r = 0.98
r = 0.99
r = 0.99
r = 0.65
r = 0.65
r = 0.96
Mean absolute updates (UP)
PE bins (1 bin = 0.1)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias


r = 0.29
r = 0.87
r = 0.93
r = 0.74
r = 0.61
r = 0.55
Mean absolute learning rate (LR)
PE bins (1 bin = 0.1)
Condiff bins (1 bin = 0.01)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias


r = 0.22
r = 0.96
r = 0.93
r = 0.9
r = 0.91
r = 0.5
Mean absolute updates (UP)
Mean absolute LR (LR)
Condiff bins (1 bin = 0.1)
Condiff bins (1 bin = 0.01)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias


r = 0.08
r = 0.95
r = 0.9
r = 0.63
r = 0.83
r = 0.89
Mean signed LR (LR)
Condiff bins (1 bin = 0.1)
Condiff bins (1 bin = 0.01)
basicRL
RLsigma
RLSigma + Confirm bias
BayesianAgent
BayesianAgent + Confirm bias